

torch.ormqr#

<https://pytorch.org/docs/stable/generated/torch.ormqr.html>

Description:

Computes the matrix-matrix multiplication of a product of Householder matrices with a general matrix. Multiplies a $m \times n$ matrix C (given by `other`) with a matrix Q , where Q is represented using Householder reflectors (`input`, `tau`). See Representation of Orthogonal or Unitary Matrices for further details. If `left` is `True` then $\text{op}(Q) \times C$ is computed, otherwise the result is $C \times \text{op}(Q)$. When `left` is `True`, the implicit matrix Q has size $m \times m$. It has size $n \times n$ otherwise. If `transpose` is `True` then op is the conjugate transpose operation, otherwise it's a no-op. Supports inputs of `float`, `double`, `cfloat` and `cdouble` dtypes. Also supports batched inputs, and, if the input is batched, the output is batched with the same dimensions.

Function Signature

```
torch.ormqr(input, tau, other, left=True, transpose=False, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output Tensor. Ignored if None. Default: None.

Notes & Warnings

NOTE

See also `torch.geqrf()` can be used to form the Householder representation (input, tau) of matrix Q from the QR decomposition.

NOTE

Note This function supports backward but it is only fast when (input, tau) do not require gradients and/or `tau.size(-1)` is very small. ``

Detailed Documentation

Documentation

Computes the matrix-matrix multiplication of a product of Householder matrices with a general matrix.

Multiplies a $m \times n$ matrix C (given by `other`) with a matrix Q , where Q is represented using Householder reflectors (`input`, `tau`). See Representation of Orthogonal or Unitary Matrices for further details.

If `left` is `True` then $op(Q)$ times C is computed, otherwise the result is C times $op(Q)$. When `left` is `True`, the implicit matrix Q has size $m \times m$. It has size $n \times n$ otherwise. If `transpose` is `True` then op is the conjugate transpose operation, otherwise it's a no-op.

Supports inputs of `float`, `double`, `cfloat` and `cdouble` dtypes. Also supports batched inputs, and, if the input is batched, the output is batched with the same dimensions.

`torch.geqrf()` can be used to form the Householder representation (`input`, `tau`) of matrix Q from the QR decomposition.

This function supports backward but it is only fast when (`input`, `tau`) do not require gradients and/or `tau.size(-1)` is very small. ``

`input` (Tensor) – tensor of shape $(*, mn, k)$ where $*$ is zero or more batch dimensions and mn equals to m or n depending on the `left`.

`tau` (Tensor) – tensor of shape $(*, \min(mn, k))$ where $*$ is zero or more batch dimensions.

`other` (Tensor) – tensor of shape $(*, m, n)$ where $*$ is zero or more batch dimensions.

left (bool) – controls the order of multiplication.

transpose (bool) – controls whether the matrix Q is conjugate transposed or not.

out (Tensor, optional) – the output Tensor. Ignored if None. Default: None.